

SHAREit Group: Facilitating digital content sharing and distribution with the power of big data

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ABOUT SHAREIT GROUP



To enrich its multi-architecture and leverage the power of big data for product development, SHAREit Group uses Google Cloud to realize more cost-effective use of cloud resources and more efficient data

About SHAREit Group

SHAREit Group is mainly dedicated to mobile internet application development and digital services. SHAREit Group has built numerous popular tools and digital entertainment content applications. The diversified suite of applications, including the core app SHAREit, has been installed by more than 2.4 billion users worldwide. SHAREit has been recognized as the "Nationally Favored App" in Southeast Asia, South Asia, Mid-East, Africa, and Russian-speaking countries. SHAREit consistently ranks

analytics, while in the connectivity and sharing service.

on the top in Google Play's Tools List in the emerging markets.

Industries: Technology

Location: Singapore

Google Cloud results

- Reduces connection disruption rate by 5% for smooth file sharing with Cloud CDN
- Helps save 40% of operational costs with autoscaling features of GKE
- Supports faster data generation with cost-effective pricing schemes through BigQuery

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Specializing

in providing cloud management services for enterprises, Cloud



Cloud**Gears**

About FunTech

FunTech's brand name is CloudGears.

CloudGears is a cloud management service provider, headquartered in Hong Kong, with branches in Beijing and Shanghai, providing enterprise customers with products, services, and solutions in the two fields of basic cloud services and big data, including digital transformation and cloud migration. Various consulting services, migration services across multi-cloud environments (including hybrid clouds), cost optimization services on the cloud, security governance services, and operation and maintenance hosting services, etc., serve global customers, covering retail, finance, manufacturing, Internet, e-commerce, games, and other industries.

with products, services, and solutions in the two fields of basic cloud services and big data, including digital transformation

- Provides accurate A/B test results for product improvement with Firebase

Google Cloud (<https://cloud.google.com/>) offering services migration services across multi-
BigQuery (<https://cloud.google.com/bigquery/>)

Cloud CDN (<https://cloud.google.com/cdn/>) the cloud security, governance services, and
Firebase (<https://firebase.google.com/>) operation and maintenance hosting

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)

Spot VMs (<https://cloud.google.com/spot-vms/>)

Digital content has become one of the most important industries. means of entertainment and communication in the world. However, distributing digital content might not be as simple as it seems due to external restrictions, like poor network, language barriers, and geographical distance.

Cloud Composer
(<https://cloud.google.com/composer/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

SHAREit Group is mainly dedicated to mobile internet application development and digital services. Upholding the corporate vision of "making digital content equally accessible by everyone" on a global level, SHAREit Group has built numerous popular tools and digital entertainment content applications. The diversified suite of applications, including the core app SHAREit (<https://play.google.com/store/apps/details?id=com.lenovo.anyshare.gps&hl=en&gl=US>), has been installed by more than 2.4 billion users worldwide. SHAREit has been recognized as the "Nationally Favored App" in Southeast Asia, South Asia, Mid-East, Africa, and Russian-speaking countries. SHAREit consistently ranks on the top in Google Play's Tools List in the emerging markets. According to the App Annie report, "Most Popular Google Play Apps of All Time," SHAREit is among the Top 10 most downloaded apps.

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*—Youze Liang, Big Data
Engineer, SHAREit Group*

Since its early days, SHAREit Group has adopted a multi-cloud strategy to deploy its products and run data analytics, because it wants to avoid provider lock-in, take advantage of the best-of-breed technologies, and prevent system failures when one public cloud platform encounters technical issues. According to Youze Liang, big data engineer at SHAREit Group, leveraging several cloud platforms also enables the company to use cloud resources in a more cost-effective way, and provide better services to end users.

"Having users in more than 200 countries means that we need to diversify our use of cloud tools to provide the best-possible services in every region. With various options at hand, we can easily design the most ideal way to deploy our products and ensure service quality and operational efficiency," he adds.

In 2019, SHAREit Group started leveraging Cloud CDN (<https://cloud.google.com/cdn>) as the network backbone of SHAREit, which has many users living in regions with poor network infrastructure. Before using Cloud CDN, these users often encountered connectivity issues when sharing files. With the extensive global network that supports Cloud CDN, SHAREit Group has enhanced its service quality by realizing a 5% lower network disruption rate.

Compared to other similar offerings, Cloud CDN has also helped the company save 10% in costs.

In early 2022, with this positive experience in mind, SHAREit Group decided to incorporate Google Cloud (<https://cloud.google.com/>) into the multi-cloud architecture of its big data platform, on which it relies for product development and business decision-making, because Google Cloud supports more cost-effective cloud usage, provides high-performance data analytics tools, and has a user interface that the SHAREit Group team is already familiar with.

"We've always been expanding the implementation of our multi-cloud strategy by adding cloud platforms that can bring greater value to our business and users," explains Liang. "Google Cloud is a perfect addition to have, as its cost-effective, high-performance infrastructure and data analytics tools can help us further unleash the power of big data."

Saving infrastructure costs with GKE and Spot VMs

SHAREit Group has been using DataCake (<https://www.datacake.cloud/login>), a cloud-based big data platform developed by FunTech, to analyze and manage its business data since 2021. With the technical support provided by FunTech on cloud product selection and infrastructure architecture design, SHAREit Group successfully merged Google Cloud and DataCake within three days. Liang says that the high level of standardization of Google

Cloud tools also helped accelerate the deployment process, because the SHAREit Group team could easily set up new authorizations and adapt its existing IT structure to Google Cloud.

SHAREit Group deploys the DataCake data platform, including Spark on Kubernetes clusters, on Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE), and automatically uses Spot VMs

(<https://cloud.google.com/spot-vms>) to support DataCake. The SHAREit Group team found that GKE not only delivers 15% higher scaling efficiency compared to other similar offerings, but also helps save operational costs by preventing computing resource waste through its autoscaling feature. Without this feature, SHAREit Group would have paid 40% more for unnecessary computing resources, or still paid nearly 20% more if it tried to scale up and down the Kubernetes clusters according to its needs on its own.

The company chose to adopt Spot VMs because Spot instances offer up to 91% lower costs than regular compute instances. However, Spot instances could be preempted, and the workloads supported by Spot VMs would need to be rerun if this happens, which increases operational costs. To fully take advantage of the low pricing of Spot VMs while ensuring operational stability, DataCake has a set of Spot utilization mechanisms, providing an elegant exit for the Spot instances recycling scenario, reducing the execution time of the same computing task by 20% to 40%, and reducing the cost by

30%-50%, allowing Spot play great value in big data scenarios.

"By providing a cost-effective cloud infrastructure for our big data platform, GKE and Spot VMs have made it possible for us to realize data-driven decision-making at very low costs," notes Liang. "We now have more resources to invest in product development and innovation."

Supporting efficient data analytics with BigQuery and Firebase

SHAREit Group leverages BigQuery

(<https://cloud.google.com/bigquery/>) as one of the query engines that support its big data platform, primarily because it offers more flexible pricing schemes than other cloud data warehouse tools. Before migrating to BigQuery, SHAREit Group used a data warehouse product that charged only based on the amount of data. As the company has many workloads that contain large amounts of data that require only simple computing, the limited pricing scheme translated into high costs. With BigQuery, SHAREit Group can choose to pay by data amount, by hour, or a monthly forfeit according to the nature of its workloads, which has greatly reduced the costs.

Besides cost-savings, BigQuery has also helped enhance the SHAREit Group team's work efficiency. The query saving and sharing feature of BigQuery facilitates the collaboration between different departments, while its capability to generate several

terabytes of data in only a few seconds accelerates the company's data processing speed.

"Adding BigQuery and Firebase to our data pipeline has boosted our product development and business decision-making. We can now process business data faster and retrieve more useful insights to develop better features for our users."

—Quan Ding, Data Analyst, SHAREit Group

For its new product development, SHAREit Group uses Firestore (<https://firebase.google.com/>) to conduct A/B tests. According to the company's comparison test, Firestore delivers the most accurate A/B test results among similar offerings. This assures the SHAREit Group team that it can make the best decisions for product development based on the A/B test results generated by Firestore. The company also has its business data in Firestore transferred directly to BigQuery for further analysis, which only takes a few minutes, in comparison to half a day if the data is transferred offline. This way, the SHAREit Group team is able to make data-driven decisions faster, and react in a more timely manner.

"Adding BigQuery and Firestore to our data pipeline has boosted our product development and business decision-making. We can now process business data faster and retrieve more useful insights to develop better features for our users," says Liang.

Offering high compatibility with open-source ecosystems

For SHAREit Group, successfully implementing its multi-cloud strategy means that it has to offer a unified user experience on all the cloud platforms it uses. To achieve this, the company adopts many open-source tools to build and run its products. Liang notes that using Google Cloud is particularly helpful for SHAREit Group to optimize its services deployed in a multi-cloud environment, because Google Cloud is highly compatible with open-source software. The company currently leverages Cloud Composer (<https://cloud.google.com/composer>), which

is built on the [Apache Airflow](https://airflow.apache.org/) (https://airflow.apache.org/) open source project, to monitor its big data workflow, and [Dataproc](https://cloud.google.com/dataproc) (https://cloud.google.com/dataproc) to run open-source data analytics tools like [Apache Spark](https://spark.apache.org/) (https://spark.apache.org/) and [Apache Flink](https://flink.apache.org/) (https://flink.apache.org/).

"After adopting Google Cloud, we've been working on standardizing our services deployed in different public clouds, which can be more easily achieved by using open-source software," says Liang. "The high compatibility of Google Cloud with open-source tools enables us to reach this goal rather effortlessly."

Leveraging AI to develop innovative features

As a part of its multi-cloud implementation plan, SHAREit Group is currently deploying its ad services powered by [Google Ad Manager](https://admanager.google.com/home/) (https://admanager.google.com/home/) on Google Cloud using [GKE, Cloud Load Balancing](https://cloud.google.com/load-balancing) (https://cloud.google.com/load-balancing), and [Compute Engine](https://cloud.google.com/compute) (https://cloud.google.com/compute). Moving forward, the company will take advantage of the artificial intelligence (AI) capabilities of Google Cloud, such as [Translation AI](https://cloud.google.com/translate) (https://cloud.google.com/translate), [Vision AI](https://cloud.google.com/vision) (https://cloud.google.com/vision), and [AutoML](https://cloud.google.com/automl) (https://cloud.google.com/automl) to develop new features. For example, it plans to use AutoML to predict users' needs based on user portraits.

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—Jianbo Zhao, Cloud
Engineer, SHAREit Group

Liang says, "The powerful infrastructure and data analytics tools of Google Cloud have helped us smoothly optimize our operations in a multi-cloud environment in a short time. We believe that we can further improve our products and bring more innovative features to users by expanding our use of Google Cloud."